Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/26/2005
PI Name:	Badler, Norman I Ph.D.		
Project Title:	RIVET: Rapid Interactive Visualization for Extensible Training		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6389	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	07/26/2004	**End Date:**	06/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Whitmore, Mihriban (NASA JSC) Megahed, Abe (Orbital Technologies Corporation)		
Grant/Contract No.:	NNJ04HF01A		
Performance Goal No.:			
Performance Goal Text:			
	The new NASA mandate calls for missions of unprecedented remoteness and duration. Challenges include high system complexity, and low training time and tolerance for error. Human capabilities remain relatively fixed and current training and instruction tools are inadequate. The RIVET team (UPenn, Orbetic, and NASA JSC) will provide computer based integrated training and instruction tools that are visually intuitive and adaptable to user skill level and context. As humans travel longer and further in space, the tasks they need to perform will increase in variability, complexity, uniqueness, and expediency, yet there will be decreasing experience, training time, and tolerance for error. Human performance failure due to inadequate accommodation of human cognitive limitations and capabilities is a major concern on future space missions. If human cognitive performance capabilities are surpassed due to inadequate design of tools, interfaces, tasks or information support systems, mission failure or decreased effectiveness or efficiency may result. Identifying, locating, processing or evaluating information to make decisions and perform critical tasks in short		

Task Description:	time-frames in nominal and emergency situations, with limited crew size, relying on strictly local resources is extremely subject to human error. This project will prototype and deliver an extensible (scalable) embedded training system (RIVET) based on graphical and speech interfaces for the user and procedural and non-programming interfaces for task training developers. RIVET is an instructional task visualization and embedded training tool that: • Is based on rapid generation of procedures by non-programming methods. • Is designed for multiple application environments (e.g., operations, maintenance, medical). • Is sympathetic to the various manners in which instructions are provided to crew today, but which permit transformations between forms to accommodate individual preferences or work conditions. • Is extensible (scalable) by instruction authors or other individuals, whether in flight or on the ground. • Supports a variety of output media including 3D virtual reality (VR), navigable animations, 2D movies, text, and speech. • Provides user interaction tools appropriate to both training and operational environments (e.g. portable systems or hands-free voice actuation). The significance of these objectives and the methodology for procedural task simulation is that crew training and instruction will be enhanced with the concomitant expectation that human errors will be reduced, infrequent tasks can be quickly refreshed, and novel problems may be tackled. If instruction and training materials can be created, visualized, and validated by individuals who need not possess deep computer programming skills, the number and skill level of people engaged in producing this material may result in cost-savings at NASA as well. The end results of this project would be: • Integrated procedure development, training, and aiding strategies and technologies. • Embedded training that is scalable, adaptive to crew information requirements, and addresses different levels of task familiarity and information needs. • Examine role of the International Procedures Viewer (IPV) and existing/developmental NASA instructional and training systems. These results will be evaluated through usability studies undertaken by the project team. Performance measures will include the amount of facilitation provided by the RIVET system that enables the performer to complete a given task in an efficient and timely manner when compared to existing instruction delivery systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Interactive instructions are not limited to only space applications. Better instructions and embedded training delivery could improve usability, accessibility, and safety for devices from home appliances to factory automation.
Task Progress:	The ACLS application was selected as the pilot instructional domain for RIVET. Several prototype interfaces were developed to show ACLS instructions in both training and instructional modes. Ultimately, a java enabled web browser was used as the base platform. This platform also enabled us to port the application to a PDA. The interface contains an area for instruction display including the current instruction, previous instructions, and future instructions dependent current instruction outcome. There are also controls for switching between performing and training modes, turning on and off speech recognition and output as well as instruction timers. A user can also restart the algorithm at anytime or undo procedures one at a time. Patient state is clearly displayed and can be altered by the user which results in jumping to the place in the algorithm that first detects the new patient state. Finally, there is a window to display images, movies, or detailed information that might accompany the current instruction. Additionally, there is a help system that includes tool tips and a help menu. As the user of our system completes instructions, they can be checked off. This helps the user to keep there place in the algorithm and enables RIVET to keep an inventory of items used as well as time taken to perform the procedures. Some of the ACLS procedures are time critical. Most instructions in the ACLS algorithm include yes or no questions that determine the next procedure that should be performed. Our system provides users the ability to look ahead to the next possible instructions before choosing an answer to the current instruction. Training mode includes sample patients and their symptoms. A user can practice procedures and algorithm navigation with the same interface that they would use in a real world application. This provides better familiarity with the system and better use of the tool.
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